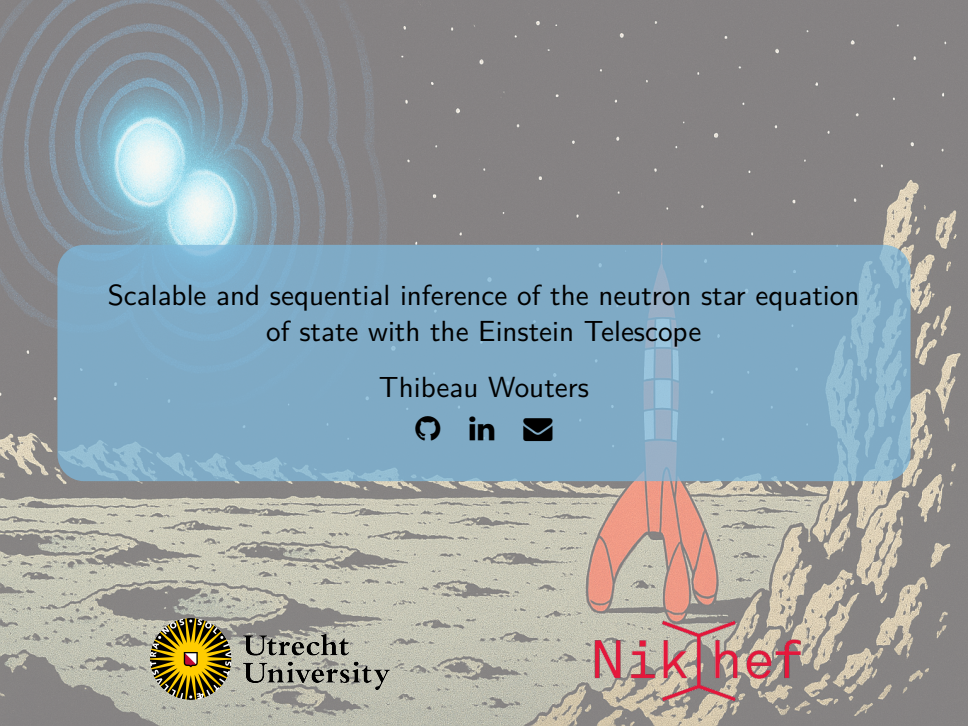




Scalable and sequential inference of the neutron star equation of state with the Einstein Telescope

ThibEAU Wouters



Utrecht
University



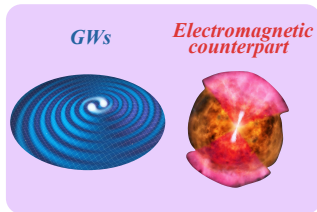
Neutron stars and fundamental physics

Neutron star mergers, especially when observed as multimessenger events (e.g., GW170817), probe

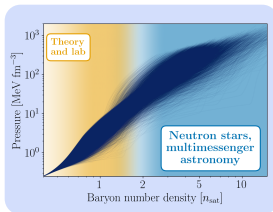
- Equation of state of ultradense matter [1]
- Expansion of the universe (H_0) [2]
- Tests of GR, populations, nucleosynthesis,...

data analysis

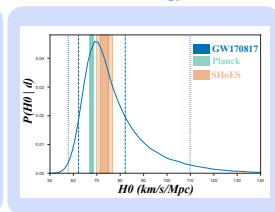
Multimessenger observations



Subatomic physics

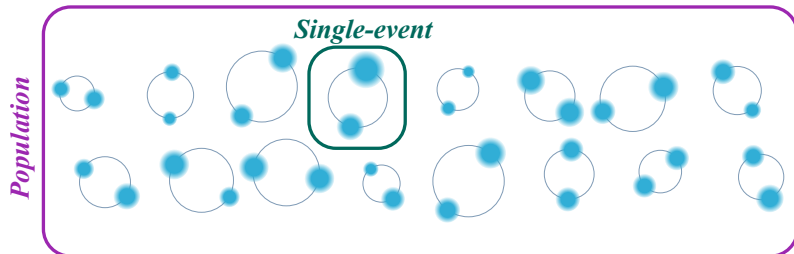


Cosmology

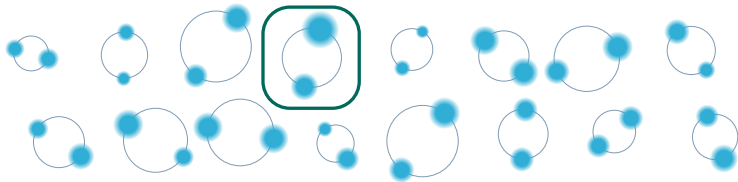


Challenges for Einstein Telescope

- Einstein Telescope
 - $\sim 30,000$ binary neutron star mergers/year
 - ~ 100 with detected electromagnetic counterpart
- **Challenge:** Data analysis (Bayesian inference) needs to scale up:
 - **Single-event:** analyze each event independently
 - **Population:** combine events into science insights
- **Solution:** GPU-acceleration and advanced algorithms



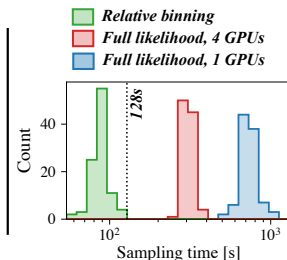
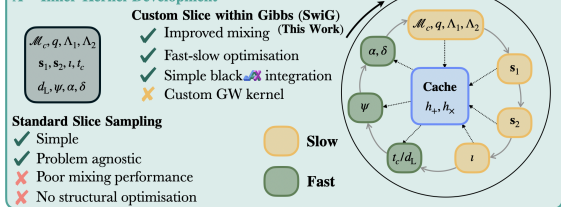
Single-event



Single event: slice-within-Gibbs sampler

- Standard sampling: move in full parameter space
- **Slice-within-Gibbs** [3]: move in ‘blocks’
 - **Slow**: new waveform generations, cached
 - **Fast**: projections, reuse cache
- Inference (for LVK signals!) in 15 mins with **no** approximation, or faster than signal duration (128 s) with relative binning

A – Inner Kernel Development



Towards ET with the BeNL community

- Development extended towards ET signals
 - RIPPLE 🌀: state-of-the-art waveforms (**talk by Robin**)
 - JIM 🌀: developing samplers (Thomas C. K. Ng)
- Strong BeNL presence (Harsh Narola, Peter T. H. Pang, Fabian Gittins, Isaac C. F. Wong, . . .)

New hardware-accelerated and accurate gravitational waveform approximants for fast parameter estimation

📅 Oct 9, 2026, 11:50 AM

🕒 20m

📍 Aula Magna

Robin Chan



RIPPLE

Thomas C. K. Ng



JIM

+ *Jef Heynen, Mick Wright,
Justin Janquart for lensing!*

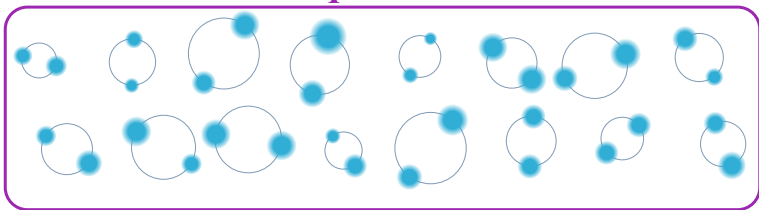
GPU-accelerated Bayesian inference for diffracted gravitational wave signals

📅 Oct 8, 2026, 5:50 PM

🕒 20m

📍 Aula Magna

Population

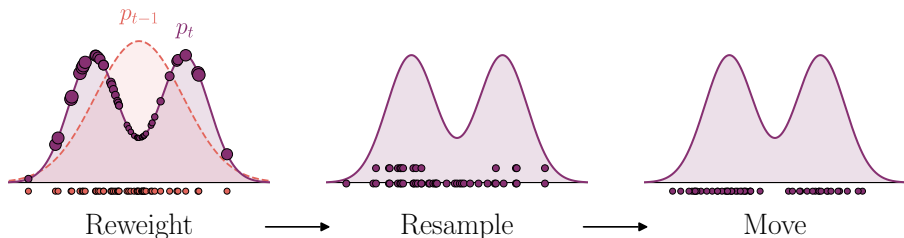


JESTER and sequential Monte Carlo (SMC)

- JESTER  [4]: used by LVK and ET
- From N events d_i to equation of state (EOS) posterior

$$\mathcal{P}(\theta_{\text{EOS}}|\{d_1, \dots, d_N\}) \propto \mathcal{L}(\{d_1, \dots, d_N\}|\theta_{\text{EOS}}) \times \pi(\theta_{\text{EOS}})$$

- **Sequential Monte Carlo**: particles from prior $\rightarrow$ posterior through a sequence of intermediate densities p_t



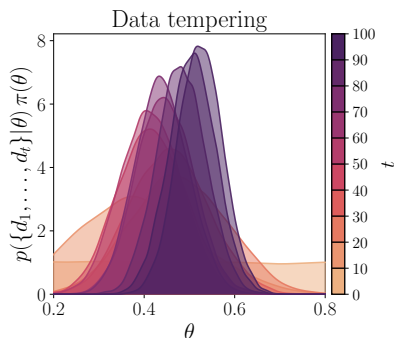
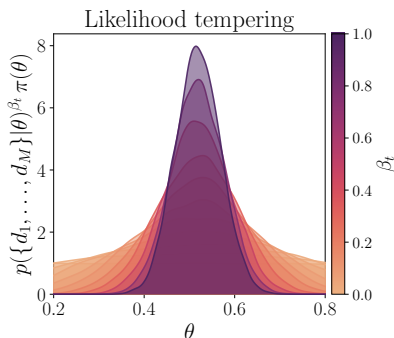
Likelihood vs. data tempering

① Likelihood tempering:

$$p_t(\theta) \propto \mathcal{L}(\{d_1, \dots, d_N\}|\theta)^{\beta_t} \pi(\theta) \quad \beta_t : 0 \rightarrow 1$$

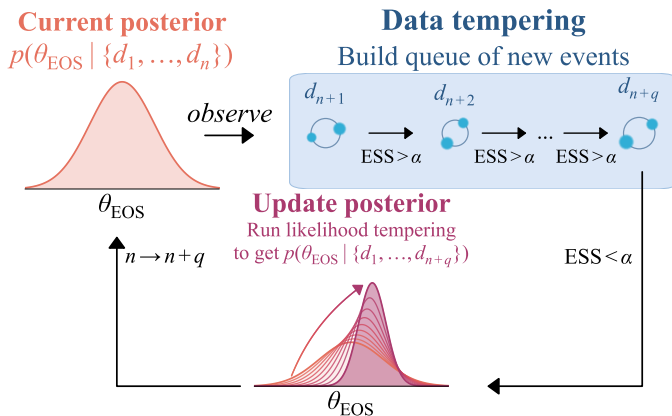
② Data tempering:

$$p_t(\theta) \propto \mathcal{L}(\{d_1, \dots, d_t\}|\theta) \pi(\theta) \quad t : 1 \rightarrow N$$



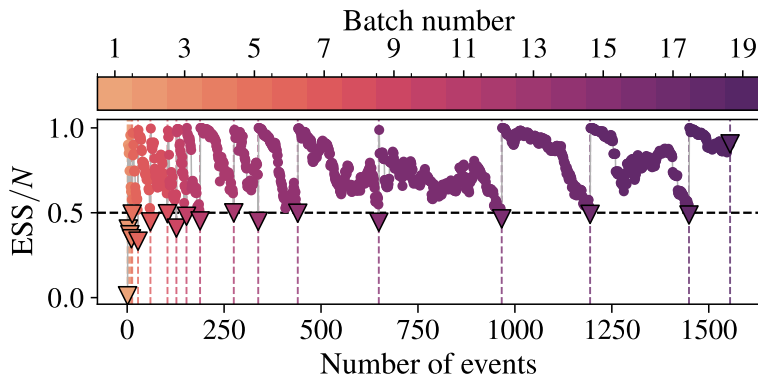
Adaptive hybrid algorithm

- ET catalogs will grow rapidly
- Need to sequentially update posteriors (data tempering)
- Use likelihood tempering for robustness



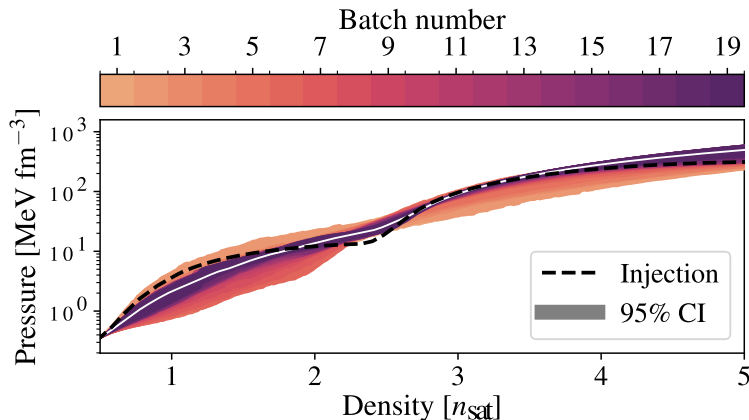
Proof of concept

- Toy catalogue: 1 month of binary neutron stars (GW) (~ 1500)
- Runtime ~ 2 hrs; expectation for 1 year: ~ 2 days



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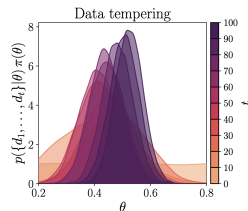
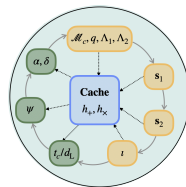
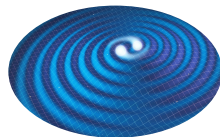


Multimessenger forecasts

- Multimessenger forecasts for population + equation of state + cosmology with JESTER [5]
- Runtime ~ 2 hrs; expectation for 1 year: ~ 2 days

Conclusions

- ET challenges data analysis for neutron stars
- GPU $\times$ software innovations
 - **Single-event**: slice-within-Gibbs
 - **Population**: sequential Monte Carlo
- Results
 - **ripple, jim**: ET under development
 - **jester**
 - Scale to $\mathcal{O}(10^3)$ sources
 - Multimessenger forecasts



Thanks for listening!



References

- [1] B. P. Abbott et al. “GW170817: Measurements of neutron star radii and equation of state”. In: *Phys. Rev. Lett.* 121.16 (2018), p. 161101. DOI: [10.1103/PhysRevLett.121.161101](https://doi.org/10.1103/PhysRevLett.121.161101). arXiv: [1805.11581](https://arxiv.org/abs/1805.11581) [gr-qc].
- [2] B. P. Abbott et al. “A gravitational-wave standard siren measurement of the Hubble constant”. In: *Nature* 551.7678 (2017), pp. 85–88. DOI: [10.1038/nature24471](https://doi.org/10.1038/nature24471). arXiv: [1710.05835](https://arxiv.org/abs/1710.05835) [astro-ph.CO].
- [3] David Yallup et al. “Ab Initio Real-Time Gravitational-Wave Parameter Estimation”. In: (July 2026). arXiv: [2607.28265](https://arxiv.org/abs/2607.28265) [gr-qc].
- [4] Thibaud Wouters et al. “Leveraging differentiable programming in the inverse problem of neutron stars”. In: *Phys. Rev. D* 112.4 (2025), p. 043037. DOI: [10.1103/v2y8-kxvx](https://doi.org/10.1103/v2y8-kxvx). arXiv: [2504.15893](https://arxiv.org/abs/2504.15893) [astro-ph.HE].
- [5] Hauke Koehn et al. “Binary neutron stars in the next-generation era: Multi-messenger detection prospects and constraints on the equation of state, mass distribution, and cosmology”. In: (July 2026). arXiv: [2607.28438](https://arxiv.org/abs/2607.28438) [astro-ph.HE].

APPENDIX